Hooking

Prehooking (Bypass Hooking) ↓

Posthooking (A ↓

(Pre-hooks ——→ WndProc) (WndProc→ defWndProc)

↓

Hooking in Post Hooking

Books:- Animal Farm (GEORGE ORWELL)

Nineteen Eighty- Four

Code:-

```
LRESULT  CALLBACK WndProc (HWND hwnd, UINT iMsg
         WPARAM wParam, LPARAM lParam)

{
     switch( iMsg )
    {
     case  WM_CREATE:
 MessageBox (hwnd, TEXT(" This is 1st Message"),
          TEXT(" WM_CREATE"), MB_OK);
          break;
    case  WM_SIZE:
 MessageBox ( hwnd, TEXT (" Window Size is Changed"),
         TEXT(" WM_SIZE"), MB_OK);
         break;
   case  WM_MOVE:
 MessageBox ( hwnd, TEXT(" Window is Moved"),
        TEXT(" WM_MOVE"), MB_OK);
         break;
   case  WM_KEYDOWN:
         switch( wParam)
          {
    case  VK_ESCAPE:
 MessageBox ( hwnd, TEXT(" Escape key is Pressed"),
       TEXT (" WM_KEYDOWN"), MB_OK);
    break;
     default:
    break;
  }
```

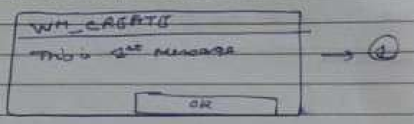
```
        case WM_CHAR:
            Switch(wParam)
        {
        case 'F':
MessageBox( hwnd, TEXT(" Capital Key 'F' is pressed ")
           TEXT(" WM_CHAR"), MB_OK);
        break;
        case 'f':
MessageBox( hwnd, TEXT(" Small Key 'F' is pressed")
           TEXT(" WM_CHAR"), MB_OK),
           break;
        }
        break;
    case WM_LBUTTONDOWN:
MessageBox( hwnd, TEXT(" LEFT MOUSE Button is Click")
        TEXT(" WM_LBUTTONDOWN"), MB_OK);
        break;
    case WM_CLOSE:
MessageBox( hwnd, TEXT(" Window is closed",
        TEXT(" WM_Close"), MB_OK),
        break;
        DestroyWindow(hwnd);
        break;

    case WM_DESTROY:
            PostQuitMessage();
            break;
    default:
        break;
    }

return( DefWindowProc(hwnd, iMsg, wParam, lParam)
```

Run

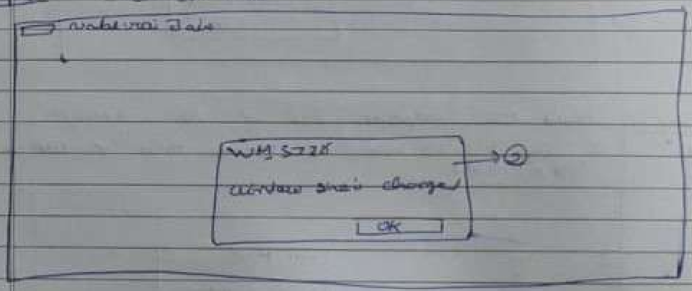
compile → cl.exe /c /EHsc Window.c
link → link.exe Window.obj User32.lib GDI32.lib

Steps To Run & Observe Output:-

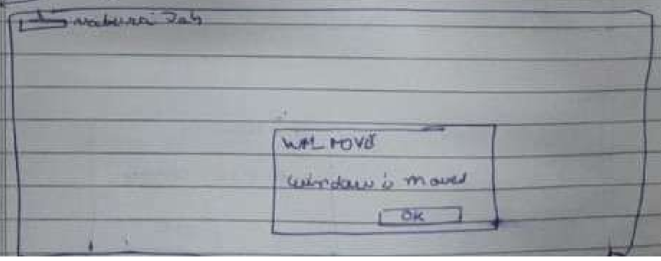
Step1:- WM_CREATE Message BOX



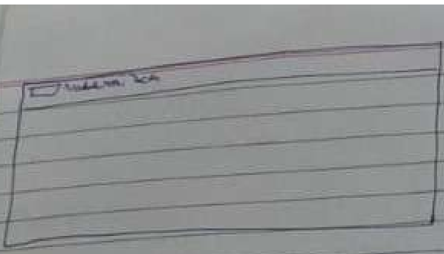
Step2:- Click on OK



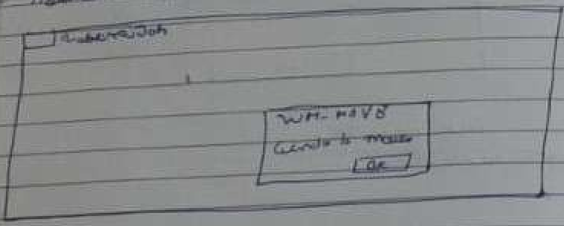
Step3 Click on OK



Step



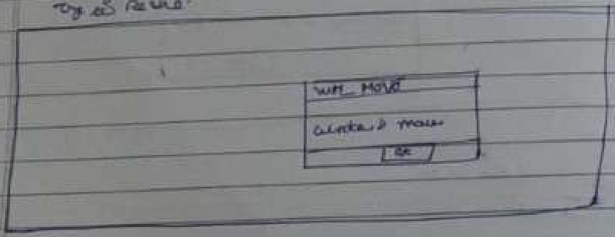
Step :- Hold the caption Bar & Move the window



WM_MOVE

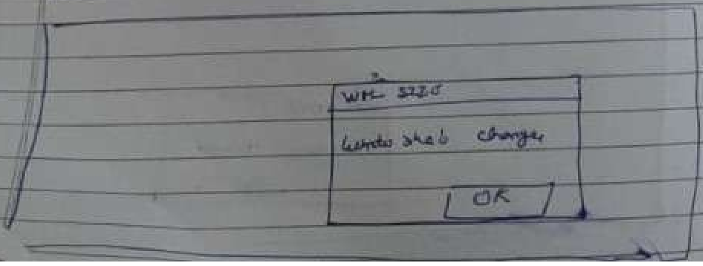
Window is moved

Step 5:- Hold the caption Bar of the window & Try to Re-size & move mouse Top, Bottom, Side, Try to Resize.



WM_MOVE

window is moved

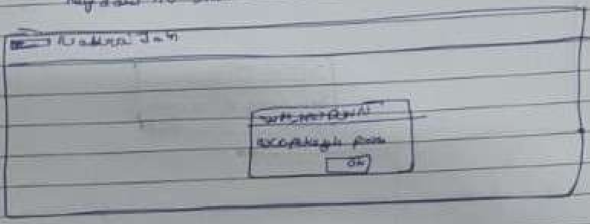


WM_SIZE

Window's size changes

Step Press Escape Key's

Keydown नहीं आली

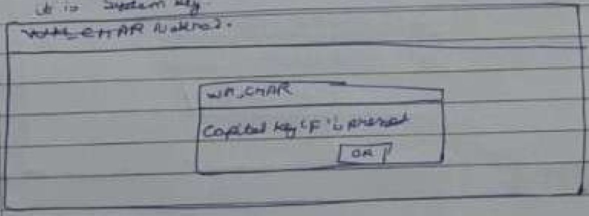


WM_KEYDOWN

Escape key is pressed

Step 7:- On the Capslock

Not Keydown because [illegible] is not a Normal key. it is System key.



WM_CHAR

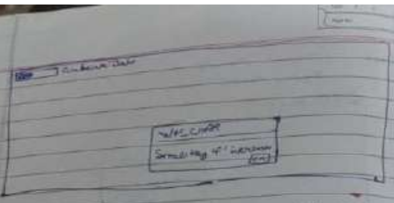
Capital key 'F' is pressed

Not any window of WM_Keydown.

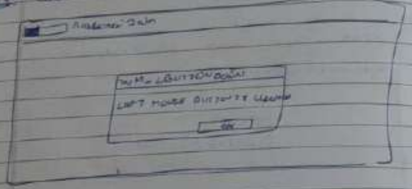
WM_Keydown OS कडे देतो, WM_CHAR कडे Translate करतो

WM_Keydown is translated into WM_CHAR if pressed key is a character key.

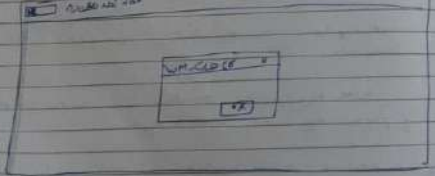
Step 8: Remove the capslock & press small f



Step 2: Clicking a mouse click on inside of the window



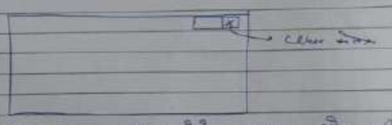
Step 3: window का (X) button दबाएँ, Msg आएगा



OK पर click करते WM_Destroy भेजेगा।

WM_DESTROY
↓
PostQuitMessage()
↓
Message Loop (WM_QUIT)
Windows (बंद होगा)

Tip: Destroy window के बाद में, Build करें।



Program अगर close होने destroy होता नहीं।
Ctrl+Alt+Del
↓
Task Manager
Windows खुलेगी।